

Phase 2: SIEM Dashboard Analysis

Objective

The goal of this phase is to collect and visualize logs from the victim machine (Metasploitable3) using a SIEM tool. We used **Splunk Enterprise** installed on the Kali attacker machine to monitor SSH brute-force attempts and identify successful attempts.

Step 1: Install Splunk Enterprise on Kali

We downloaded and installed Splunk Enterprise on the Kali machine using the official .deb package.

Command to install Splunk:

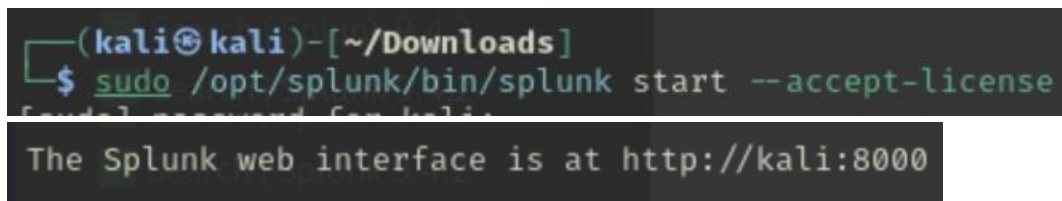
```
sudo dpkg -i splunk-9.4.2-e9664af3d956-linux-amd64.deb
```

Command to start Splunk and accept the license:

```
sudo /opt/splunk/bin/splunk start --accept-license
```

During setup, Splunk required creating an administrator account:

- Username: kali
- Password: kalikali



```
(kali@kali)-[~/Downloads]
$ sudo /opt/splunk/bin/splunk start --accept-license
[failed] account for kali
The Splunk web interface is at http://kali:8000
```

Step 2: Install Splunk Universal Forwarder on Metasploitable3

We installed the Splunk Universal Forwarder on the victim machine to send logs to the Splunk server.

Command to download and extract the forwarder (Logs were so long so no screenshots here):

```
wget -O splunkforwarder-9.4.2-e9664af3d956-linux-amd64.tgz
```

```
"https://download.splunk.com/products/universalforwarder/releases/9.4.2/linux/splunkforwarder-9.4.2-e9664af3d956-linux-amd64.tgz"
```

```
tar -xvzf splunkforwarder-9.4.2-e9664af3d956-linux-amd64.tgz
```

```
cd splunkforwarder/bin
```

Command to start the forwarder:

We created an administrator account:

- Username: kali
- Password: kalikali

Step 3: Configure the Forwarder to Send Data to Splunk Server

We connected the victim machine to the Splunk server (Kali machine) using port 9997.

Command to add the forward-server:

```
sudo ./splunk add forward-server 192.168.56.101:9997 -auth kali:kalikali
```

```
vagrant@metasploitable3-ub1404:~/splunkforwarder/bin$ sudo ./splunk add forward-server 192.168.56.101:9997 -auth kali:kalikali
```

Forwarding to port 9997 successfully

```
Added forwarding to: 192.168.56.101:9997.
```

And in Splunk Dashboard set the port setting to 9997 and Listen for it:

Listen on this port ▾	Status ▾	Actions
9997	Enabled Disable	Delete

Step 4: Repeating the Brute-Force attack after the splunk setup:

```
msf6 > use auxiliary/scanner/ssh/ssh_login
msf6 auxiliary(scanner/ssh/ssh_login) > set USER_FILE /home/kali/Desktop/users.txt
USER_FILE => /home/kali/Desktop/users.txt
msf6 auxiliary(scanner/ssh/ssh_login) > set PASS_FILE /home/kali/Desktop/passwords.txt
PASS_FILE => /home/kali/Desktop/passwords.txt
msf6 auxiliary(scanner/ssh/ssh_login) > set RHOSTS 192.168.56.102
RHOSTS => 192.168.56.102
msf6 auxiliary(scanner/ssh/ssh_login) > set VERBOSE true
VERBOSE => true
msf6 auxiliary(scanner/ssh/ssh_login) > set THREADS 4
THREADS => 4
msf6 auxiliary(scanner/ssh/ssh_login) > run
```

Step 5: Verify Log Ingestion in Splunk Dashboard

We accessed the Splunk web dashboard at:

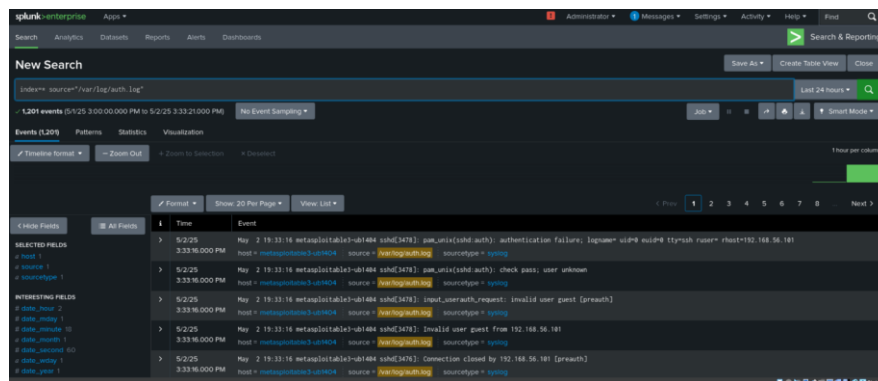
<http://kali:8000> (Had to turn off FireFox proxy -> set to no proxy)

Logged in with:

- Username: kali
- Password: kalikali

We navigated to **Search & Reporting**, then used the following search query to verify incoming logs:

Search query to confirm log ingestion:



>	5/2/25 3:33:14:00 PM	May 2 19:33:14 metasploit3-ub1404 sshd[3474]: Connection closed by 192.168.56.101 [preauth]	host = metasploit3-ub1404	source = /var/log/auth.log	sourcetype = syslog
>	5/2/25 3:33:14:00 PM	May 2 19:33:14 metasploit3-ub1404 sshd[3474]: Failed password for invalid user guest from 192.168.56.101 port 41647 ssh2	host = metasploit3-ub1404	source = /var/log/auth.log	sourcetype = syslog
>	5/2/25 3:33:12:00 PM	May 2 19:33:12 metasploit3-ub1404 sshd[3474]: pam_unix(sshd:auth): authentication failure; logname=uid=0 euid=0 tty=ssh ruser= rhost=192.168.56.101	host = metasploit3-ub1404	source = /var/log/auth.log	sourcetype = syslog
>	5/2/25 3:33:12:00 PM	May 2 19:33:12 metasploit3-ub1404 sshd[3474]: pam_unix(sshd:auth): check pass; user unknown	host = metasploit3-ub1404	source = /var/log/auth.log	sourcetype = syslog
>	5/2/25 3:33:12:00 PM	May 2 19:33:12 metasploit3-ub1404 sshd[3474]: input_userauth_request: invalid user guest [preauth]	host = metasploit3-ub1404	source = /var/log/auth.log	sourcetype = syslog
>	5/2/25 3:33:12:00 PM	May 2 19:33:12 metasploit3-ub1404 sshd[3474]: Invalid user guest from 192.168.56.101	host = metasploit3-ub1404	source = /var/log/auth.log	sourcetype = syslog
>	5/2/25 3:33:12:00 PM	May 2 19:33:12 metasploit3-ub1404 sshd[3472]: Connection closed by 192.168.56.101 [preauth]	host = metasploit3-ub1404	source = /var/log/auth.log	sourcetype = syslog
>	5/2/25 3:33:12:00 PM	May 2 19:33:12 metasploit3-ub1404 sshd[3472]: Failed password for invalid user guest from 192.168.56.101 port 43779 ssh2	host = metasploit3-ub1404	source = /var/log/auth.log	sourcetype = syslog
>	5/2/25 3:33:10:00 PM	May 2 19:33:10 metasploit3-ub1404 sshd[3472]: pam_unix(sshd:auth): authentication failure; logname=uid=0 euid=0 tty=ssh ruser= rhost=192.168.56.101	host = metasploit3-ub1404	source = /var/log/auth.log	sourcetype = syslog

Step 6: Visualize SSH Login Attempts

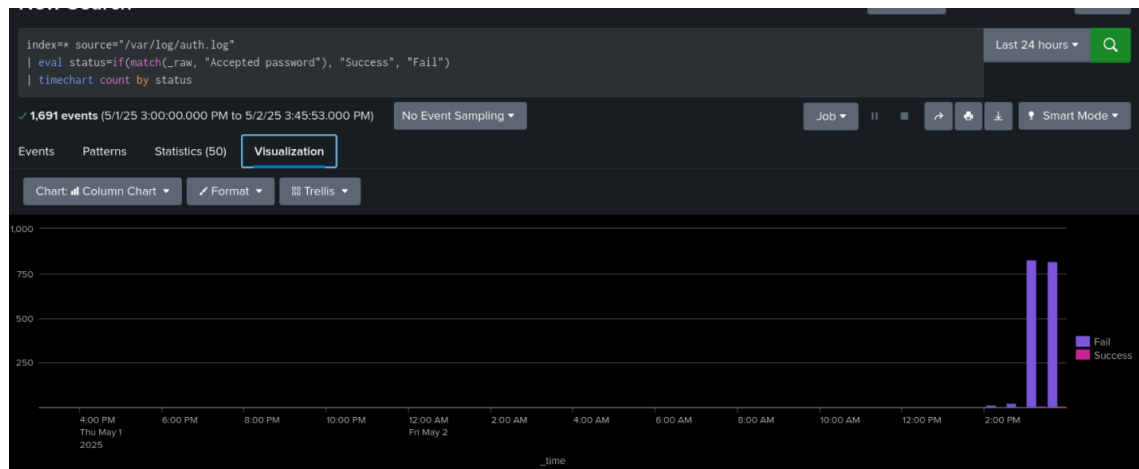
We created a chart to show the count of successful and failed login attempts.

Timechart query for login status:

`index=* source="/var/log/auth.log"`

`| eval status=if(match(_raw, "Accepted password"), "Success", "Fail")`

| timechart count by status



Step 7: Filter and Display Only Successful Logins

To isolate and analyze only the successful login events:

Search query for successful logins Or I clicked on Success from the Chart:

`index=* source="/var/log/auth.log"`

`| eval status=if(match(_raw, "Accepted password"), "Success", "Fail")`

`| search status=Success`

The image shows the Splunk search results interface. The search bar contains the query: `index=* source="/var/log/auth.log" | eval status=if(match(_raw, "Accepted password"), "Success", "Fail") | search status=Success`. The interface indicates 2 events from 5/1/25 3:00:00.000 PM to 5/2/25 3:46:59.000 PM. The results are displayed in a table format. The table has columns for 'Time' and 'Event'. The 'Event' column contains detailed log messages.

i	Time	Event
>	5/2/25 3:35:32.000 PM	May 2 19:35:32 metasploitable3-ub1404 sshd[3657]: Accepted password for vagrant from 192.168.56.101 port 35113 ssh2 host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
>	5/2/25 3:28:15.000 PM	May 2 19:28:15 metasploitable3-ub1404 sshd[3006]: Accepted password for vagrant from 192.168.56.101 port 34865 ssh2 host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog

Conclusion

In Phase 2, we achieved the following:

- Installed and configured Splunk Enterprise and Universal Forwarder
- Connected the victim (Metasploitable3) to the SIEM environment

- Forwarded real-time log data for analysis
- Visualized attack patterns and extracted insights about brute-force login attempts